

07

System of Linear Equations

Easier to Understand from the Perspective of Lines and Planes

7.1 System of Two Linear Equations



- ▶ Systems of linear equations: converting problems such as the chicken-and-rabbit puzzle into a system of two linear equations.
- ▶ Solving algebraically: finding the unknowns via substitution/elimination and the inverse-matrix method.
- ▶ Geometric perspective: each linear equation in two variables represents a line, and the solution is the intersection of the lines.
- ▶ Types of solutions: using the graph to judge whether the system has a unique solution, no solution, or infinitely many solutions.
- ▶ Expressing the system in matrix form: writing the system of linear equations as the matrix-multiplication form $A\mathbf{x} = \mathbf{b}$, which is compact and convenient for computation.
- ▶ Using whether the determinant is zero to judge whether a unique solution exists.
- ▶ Understanding elementary row operations: row swapping, row scaling, and row sum.
- ▶ Using functions such as `numpy.linalg.inv()` and `numpy.linalg.solve()` to solve systems of linear equations.

This section introduces how to solve the chicken-and-rabbit-in-a-cage problem using a system of linear equations, and analyzes its solution from both algebraic and geometric perspectives. We first set up a system of two linear equations and convert it into matrix-multiplication form. We then use the inverse-matrix method to compute the solution and verify that it makes sense.

Next, we explain linear equations in two variables from a geometric perspective, pointing out that each equation represents a line and the solution is the intersection of two lines. We explore the possible types of solutions a system can have under different conditions — a unique solution, no solution, or infinitely many solutions — and use contour plots to vividly illustrate the geometric properties involved. Finally, we discuss how different types of matrices (identity matrices, diagonal matrices, upper triangular matrices, and lower triangular matrices) affect how the system is solved.

The Chicken-and-Rabbit-in-a-Cage Problem

The Sunzi Suanjing (Master Sun's Mathematical Manual) states the chicken-and-rabbit-in-a-cage problem as follows: "There are pheasants and rabbits in the same cage. Above, there are thirty-five heads; below, there are ninety-four feet. How many pheasants and how many rabbits are there?"

We can construct two linear equations in two variables to describe this problem.

A linear equation in two variables is an algebraic equation containing two unknowns, each raised to a power no higher than 1; its general form is:

$$a_1x_1 + a_2x_2 = b$$

Here, x_1 and x_2 are the two unknowns; a_1 and a_2 are the coefficients (a_1 and a_2 can be any real numbers); and b is a constant (any real number).

Suppose the number of chickens is x_1 and the number of rabbits is x_2 .

The total number of chickens and rabbits is 35, which can be written as the linear equation

$$x_1 + x_2 = 35$$

Chickens have 2 feet and rabbits have 4 feet, for a total of 94 feet, which can be written as the linear equation

$$2x_1 + 4x_2 = 94$$

Combining the two linear equations above, the chicken-and-rabbit-in-a-cage problem can be written as the system of linear equations

$$\begin{cases} x_1 + x_2 = 35 \\ 2x_1 + 4x_2 = 94 \end{cases}$$

A system of linear equations is a set of equations made up of multiple linear equations.

Solving It

Solving this system of linear equations is easy.

Dividing both sides by 2 gives

$$x_1 + 2x_2 = 47$$

Equation — subtract from its left- and right-hand sides the left- and right-hand sides of respectively, to get

$$x_1 + 2x_2 - (x_1 + x_2) = 47 - 35$$

we get

$$x_2 = 12$$

Substituting into and simplifying gives

$$x_1 = 23$$

We find that there are 23 chickens and 12 rabbits in the cage, that is,

$$\begin{cases} x_1 = 23 \\ x_2 = 12 \end{cases}$$

Geometric Perspective

From a geometric perspective, the following linear equation in two variables represents a line

$$a_1x_1 + a_2x_2 = b$$

Different combinations of coefficients affect the shape of the line; let's look at a few cases.

When $a_2 = 0$, the equation becomes

$$a_1x_1 = b$$

As shown in Figure 1 (a), the equation above represents a vertical line. For example, $x_1 = -1$, $x_1 = 2$.

When $a_1 = 0$, the equation becomes

$$a_2x_2 = b$$

As shown in Figure 1 (b), the equation above represents a horizontal line. For example, $x_2 = 1$, $x_2 = -2$.

As shown in Figure 1 (c), when a_1 and a_2 are both nonzero, is a line with nonzero slope.

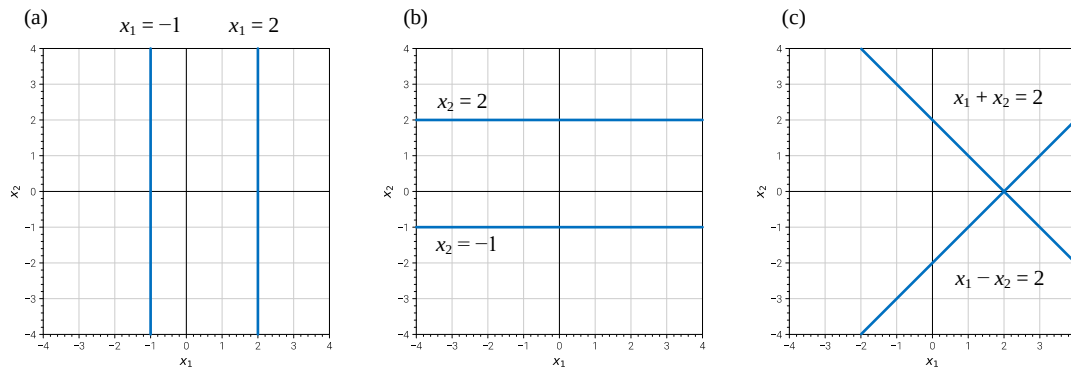


Figure 1. A Geometric View of Linear Equations in Two Variables

Solving the Chicken-and-Rabbit Problem Graphically

With the geometric perspective, we can solve the chicken-and-rabbit problem graphically.

As shown in Figure 2, the red line represents $x_1 + x_2 = 35$, meaning that every point on the red line satisfies $x_1 + x_2 = 35$.

The blue line represents $2x_1 + 4x_2 = 94$.

The two unknowns correspond to two linear equations; the number of unknowns equals the number of equations. The intersection of the red and blue lines is the solution to the chicken-and-rabbit problem.

⚠️ Note that the chicken-and-rabbit problem has one more implicit condition: the number of chickens x_1 and the number of rabbits x_2 must both be non-negative integers.

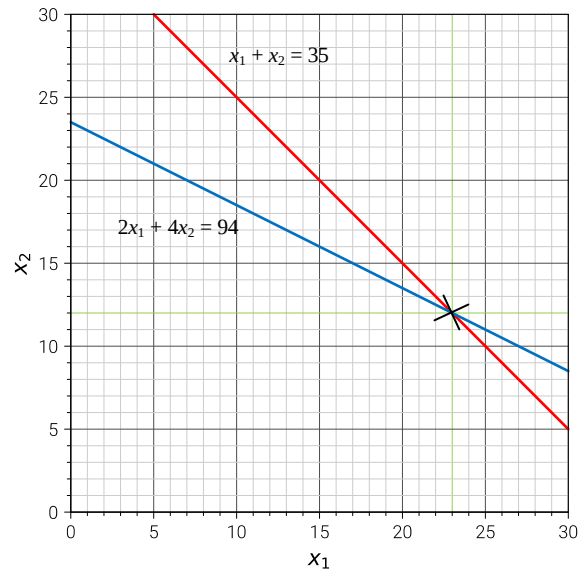


Figure 2. A Graphical Solution to the Chicken-and-Rabbit Problem

Writing It in Matrix-Multiplication Form

As shown in Figure 3 (a), writing the first equation in matrix-operation form gives

$$1 \cdot x_1 + 1 \cdot x_2 = 35 \Rightarrow \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 35 \end{bmatrix}$$

As shown in Figure 3 (b), the second equation in is written in a similar form

$$2 \cdot x_1 + 4 \cdot x_2 = 94 \Rightarrow \begin{bmatrix} 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 94 \end{bmatrix}$$

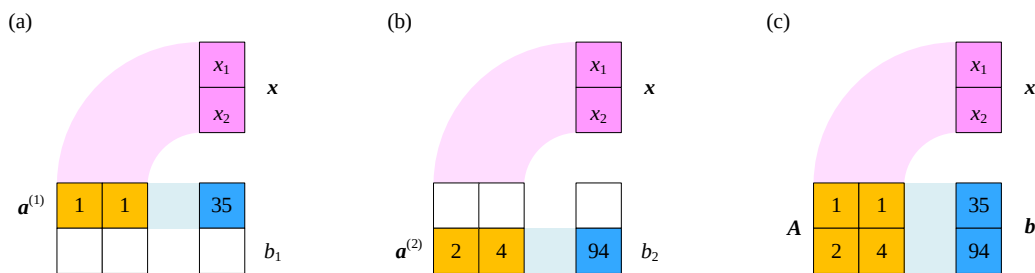


Figure 3. Writing a System of Linear Equations in Matrix-Multiplication Form

As shown in Figure 3 (c), combining and , we can write the system of linear equations for the chicken-and-rabbit problem in matrix form

$$\begin{cases} 1 \cdot x_1 + 1 \cdot x_2 = 35 \\ 2 \cdot x_1 + 4 \cdot x_2 = 94 \end{cases} \Rightarrow \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 35 \\ 94 \end{bmatrix}$$

can be written as

$$Ax = b$$

where,

$$\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 35 \\ 94 \end{bmatrix}$$

the matrix $\mathbf{A}$ is called the coefficient matrix, and $\mathbf{b}$ is called the vector of constant terms.

$\mathbf{x}$ is the column vector of unknown variables; since $\mathbf{A}$ is a square, invertible matrix, $\mathbf{x}$ can be found using the equation below

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

Computing the determinant of 's square matrix $\mathbf{A}$: $\det(\mathbf{A}) = 2$; so the matrix $\mathbf{A}$ is invertible.

Substituting in the specific numerical values and computing gives $\mathbf{x}$

$$\mathbf{x} = \begin{bmatrix} 1 & 1 \\ 2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 35 \\ 94 \end{bmatrix} = \begin{bmatrix} 2 & -0.5 \\ -1 & 0.5 \end{bmatrix} \begin{bmatrix} 35 \\ 94 \end{bmatrix} = \begin{bmatrix} 23 \\ 12 \end{bmatrix}$$

Code 1 computes the inverse of the matrix $\mathbf{A}$, and then solves the system of linear equations.

Let's go over the key statements in Code 1 below.

- a** Creates a 2D array representing the coefficient matrix in the chicken-and-rabbit problem. Each column of $\mathbf{A}$ corresponds to a different unknown, such as the number of chickens and rabbits, and each row corresponds to one equation.
- b** Creates a 2D array representing the vector of constant terms in the chicken-and-rabbit problem. The two elements of the array represent the number of heads and the number of feet, respectively.
- c** Uses `numpy.linalg.inv(A)` to compute the inverse of $\mathbf{A}$.
Please add a line before this one that uses `numpy.linalg.det()` to check whether the matrix $\mathbf{A}$ is invertible.
- d** These two lines verify that the product of the matrix and its inverse, and the product of the inverse and the matrix, are both the identity matrix.
- e** The result computed by `A_inv @ b` is $\mathbf{x}$, the final answer to the problem, which includes the number of chickens and the number of rabbits.

Code 1. Solving a System of Linear Equations Using the Inverse Matrix |  LA_Ch07_01_01.ipynb

```

## 初始化
import numpy as np

## 鸡兔同笼系数矩阵
a A = np.array([[1, 1],
                [2, 4]])

## 常数列向量
b b = np.array([[35],
                [94]])

## 矩阵A的逆
c A_inv = np.linalg.inv(A)

# 验证
d A_inv @ A
  A @ A_inv

## 求解
e A_inv @ b

```



LA_07_01_02.ipynb uses `numpy.linalg.solve()` to solve the chicken-and-rabbit problem, and LA_07_01_03.ipynb solves it using SymPy; please study these on your own.

You may be curious: why express a system of linear equations in matrix-multiplication form?

Expressing a system of linear equations in matrix-multiplication form has several advantages. First, it makes writing the system much more compact, avoiding the need to repeatedly list all the variables — this simplifies the expression significantly, especially when there are many variables. Second, the properties of the matrix provide important information; for example, by analyzing matrix A 's rank, we can quickly determine whether the system has a unique solution, no solution, or infinitely many solutions. In addition, matrix representations extend naturally to more complex linear-algebra problems, such as solving overdetermined systems, eigenvalue decomposition, and linear transformations. Finally, in computer computation, matrix operations are more efficient than solving equations one by one, substantially improving both computational speed and precision.

Two Linear Equations in Two Variables

Given two linear equations in two variables, we can write

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 = b_2 \end{cases}$$

Writing this in matrix-multiplication form:

$$\begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

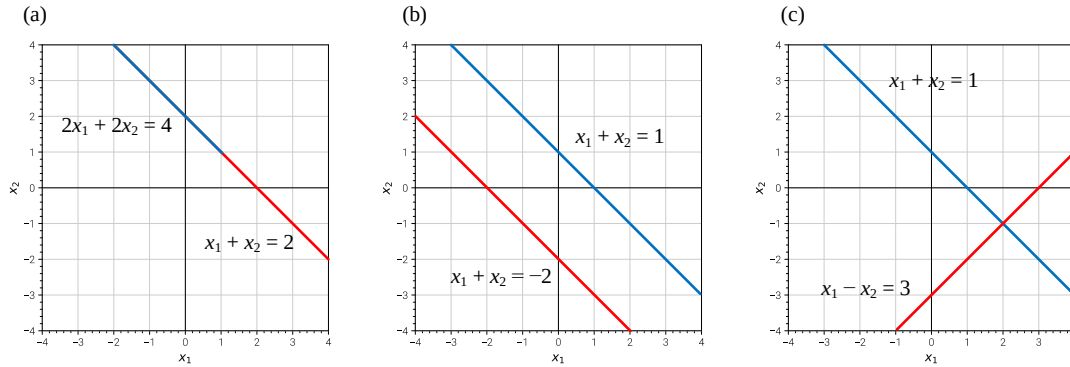


Figure 4. Different Numbers of Solutions

As shown in Figure 4 (a), the two lines completely coincide, so the system has infinitely many solutions, that is,

$$\begin{cases} x_1 + x_2 = 2 \\ 2x_1 + 2x_2 = 4 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 2 \\ 4 \end{bmatrix}}_{\mathbf{b}}$$

In fact, this is the same line, so there are infinitely many solutions. The determinant of the matrix $\mathbf{A}$ in the equation above, $\det(\mathbf{A}) = 0$.

As shown in Figure 4 (b), the two lines are parallel, so the system has no solution, that is,

$$\begin{cases} x_1 + x_2 = 1 \\ x_1 + x_2 = 3 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{\mathbf{b}}$$

The two lines have no intersection. We find that the determinant of the matrix $\mathbf{A}$ in the equation above is also 0, that is, $\det(\mathbf{A}) = 0$.

As shown in Figure 4 (c), the following two lines intersect, giving one solution

$$\begin{cases} x_1 + x_2 = 1 \\ x_1 - x_2 = 3 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{\mathbf{b}}$$

The determinant of the matrix $\mathbf{A}$ in the equation above is -2 , that is, $\det(\mathbf{A}) = -2$; this shows that the matrix $\mathbf{A}$ is invertible.

Clearly, the matrix $\mathbf{A}$ affects the result. Let's now go over the graphs corresponding to different forms of the 2×2 matrix $\mathbf{A}$.

The Identity Matrix

If the matrix $\mathbf{A}$ is the identity matrix, for example,

$$\begin{cases} x_1 = 1 \\ x_2 = -1 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_b$$

The inverse of the identity matrix is itself.

We find that the identity matrix is the “most perfect” of all, because no extra computation is needed — the column vector $\mathbf{b}$ is itself the solution to the system of linear equations, that is,

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ -1 \end{bmatrix}}_b$$

In other words, when solving $\mathbf{Ax} = \mathbf{b}$, if we can reduce the matrix $\mathbf{A}$ to the identity matrix, the system is solved.

As shown in Figure 5, $x_1 = 1$ corresponds to the vertical line (red) in the figure, and $x_2 = -1$ corresponds to the horizontal line (blue) in the figure. Their intersection is the solution.

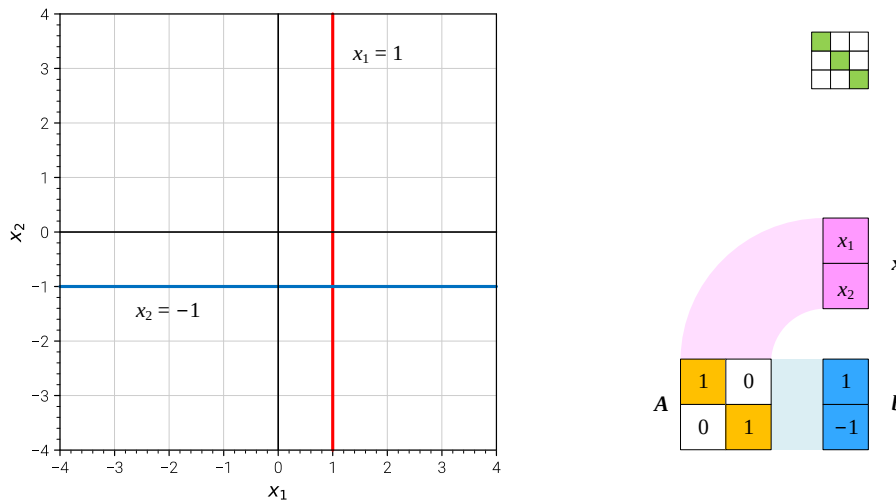


Figure 5. The Matrix $\mathbf{A}$ Is the Identity Matrix

This matches what we would expect. Earlier in this section, when we solved the chicken-and-rabbit problem, the result could be written as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 23 \\ 12 \end{bmatrix} = \begin{bmatrix} 23 \\ 12 \end{bmatrix}$$

The two equations above also correspond to a vertical line and a horizontal line, respectively.

Swapping the rows of gives

$$\begin{cases} x_2 = -1 \\ x_1 = 1 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_b$$

Figure 6 shows the graphs corresponding to the two equations above.

Clearly, swapping the rows (row swapping) of the equations does not affect the result. This is part of the rules of elementary row operations.

Elementary row operations are basic operations performed on the rows of a matrix, including row swapping, row scaling, and row sum. Elementary row operations are used to simplify a matrix through its rows, and are especially common when solving systems of linear equations, computing the rank of a matrix, and finding matrix inverses.

The matrix A is a permutation matrix; left-multiplying x by it performs a “row swap.” The determinant of this matrix is -1 .

's first row is denoted R_1 and its second row is denoted R_2 ; swapping these two rows is written $R_1 \leftrightarrow R_2$.

You may have already noticed that itself is also directly the solution of the system, but it is not as “natural” as .

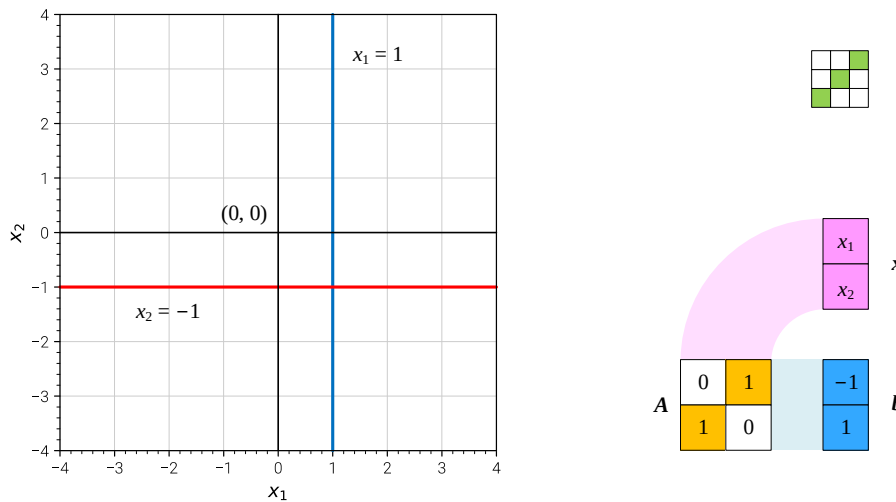


Figure 6. The Matrix A Is a Permutation Matrix

As shown in the figure below, row swapping only changes the order in which the lines are displayed.

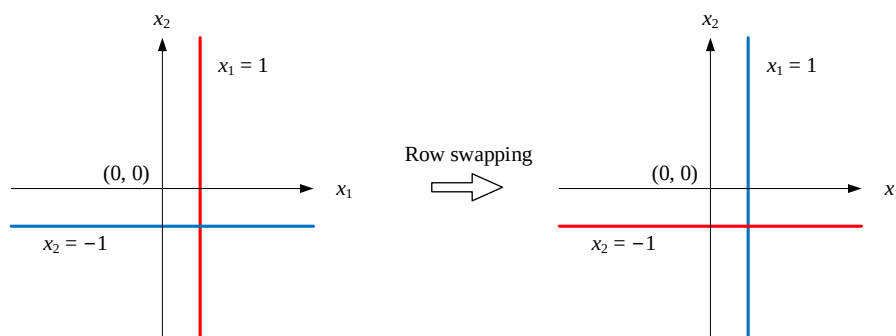


Figure 7. Row Swapping

Diagonal Matrices

If the matrix A is a diagonal matrix, for example,

$$\begin{cases} 2x_1 = 2 \\ 3x_2 = -3 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 2 \\ -3 \end{bmatrix}}_b$$

The determinant of this matrix is 6, so the matrix is invertible.

To solve x , we need to find the inverse of the diagonal matrix

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 2 \\ -3 \end{bmatrix}}_b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

As shown in Figure 8, $2x_1 = 2$ corresponds to the vertical line (red), and $3x_2 = -3$ corresponds to the horizontal line (blue). Their intersection is the solution.

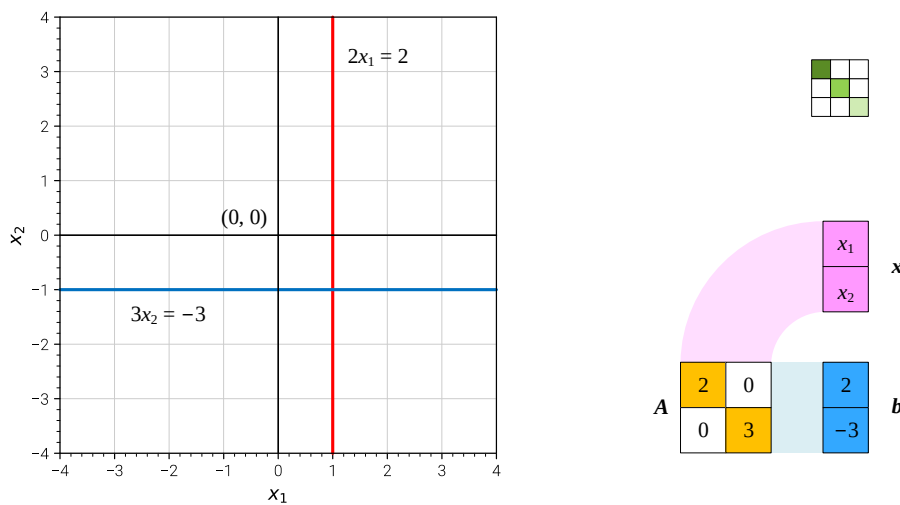


Figure 8. The Matrix A Is a Diagonal Matrix

In fact, is already very close to the solution of the system of two linear equations. We only need to apply row scaling to each equation – multiplying all elements of a row by a nonzero scalar – to obtain the solution.

As shown in the figure below, row scaling does not change the position or shape of the lines.

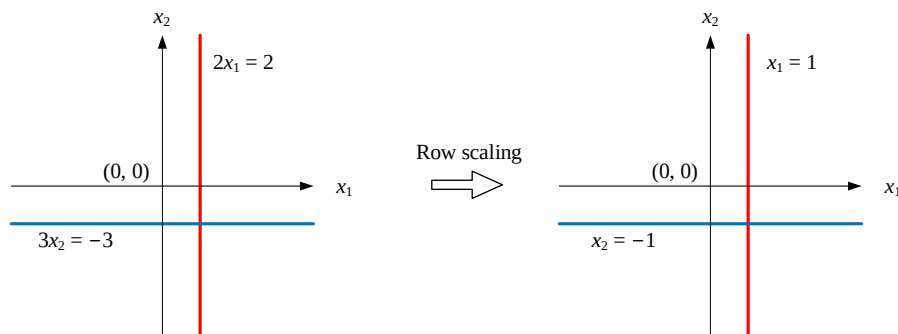


Figure 9. Row Scaling

's first row divided by 2 and second row divided by 3 gives

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

We denote these two steps as $R_1/2 \rightarrow R_1$, $R_2/3 \rightarrow R_2$. And these two operations are also hidden in 's inverse matrix A^{-1} ! We can see $1/2$ and $1/3$ in the inverse matrix A^{-1} .

In other words, even though A is already a diagonal matrix, we still need to reduce A to the identity matrix.

Upper Triangular Matrices

If the matrix A is an upper triangular matrix, for example,

$$\begin{cases} x_1 + x_2 = 0 \\ x_2 = -1 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_b$$

The determinant of this matrix is 1, so an inverse exists.

To solve x , we need to find the inverse of the upper triangular matrix

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 0 \\ -1 \end{bmatrix}}_b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

As shown in Figure 10, $x_1 + x_2 = 0$ corresponds to the diagonal line (red), and $x_2 = -1$ corresponds to the horizontal line (blue). Their intersection is the solution.

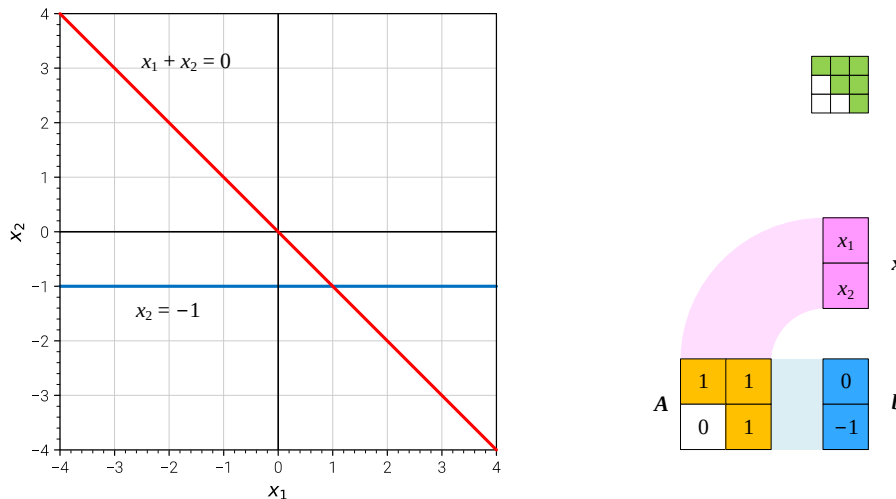


Figure 10. The Matrix A Is an Upper Triangular Matrix

Looking at , from the second row R_2 , we already know the solution for x_2 ; the first row R_1 minus the second row R_2 , used to replace R_1 , gives the solution for x_1 . This process is written $R_1 - R_2 \rightarrow R_1$. This operation is called row sum (adding a nonzero multiple of one row to another). This operation is in fact also present in 's inverse matrix A^{-1} .

This process also converts the upper triangular matrix into the identity matrix!

As shown in the figure below, row sum changes the shape and position of the lines.

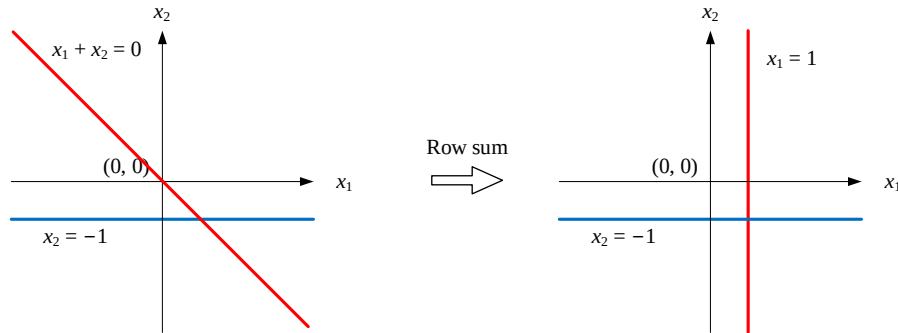


Figure 11. Row Sum

Lower Triangular Matrices

If the matrix A is a lower triangular matrix, for example,

$$\begin{cases} x_1 = 1 \\ x_1 + x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_b$$

The determinant of the matrix A above is 1, so the matrix has an inverse.

To solve x , we need to find the inverse of the lower triangular matrix

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

As shown in Figure 12, $x_1 = 1$ corresponds to the vertical line (red), and $x_1 + x_2 = 0$ corresponds to the diagonal line (blue). Their intersection is the solution.

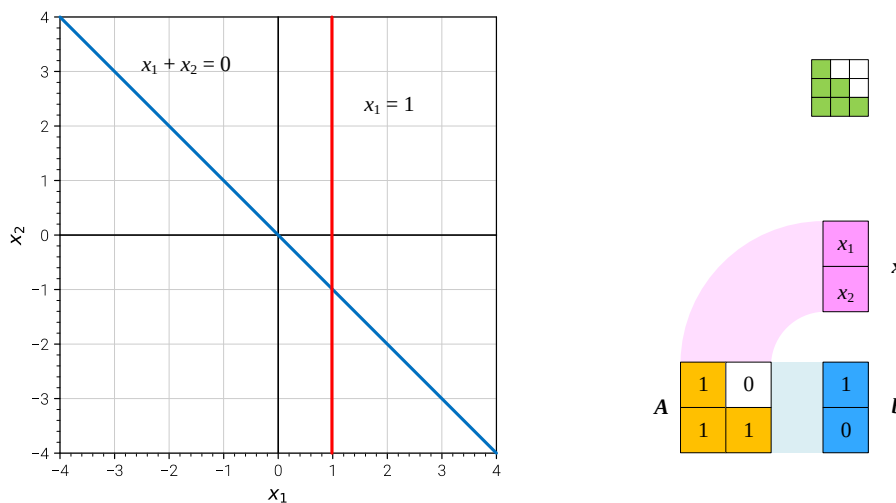


Figure 12. The Matrix A Is a Lower Triangular Matrix

Looking at , the first row R_1 gives the solution for x_1 ; the second row R_2 minus the first row R_1 , used to replace R_2 , gives the solution for x_2 . This process is written $R_2 - R_1 \rightarrow R_2$. Clearly, this process also converts the lower triangular matrix into the identity matrix!

General Square Matrices with Nonzero Determinant

If the matrix A is a general square matrix, and its determinant is nonzero, then A is invertible and the system has a unique solution — for example, the following.

$$\begin{cases} x_1 - x_2 = 2 \\ 2x_1 + 2x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 2 \\ 0 \end{bmatrix}}_b$$

The determinant of the matrix A above is 4, so the matrix has an inverse.

To solve x , we need to find the inverse of the matrix

$$\underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0.5 & 0.25 \\ -0.5 & 0.25 \end{bmatrix}}_{A^{-1}} \underbrace{\begin{bmatrix} 2 \\ 0 \end{bmatrix}}_b = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

As shown in Figure 13, $x_1 - x_2 = 2$ corresponds to the red diagonal line, and $2x_1 + 2x_2 = 0$ corresponds to the blue diagonal line. Their intersection is the solution.

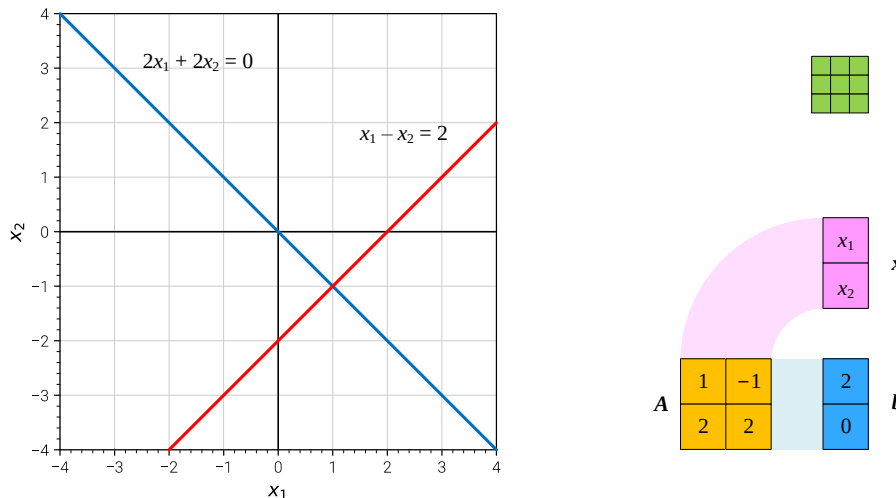


Figure 13. The Matrix A Has Nonzero Determinant, $Ax = b$



Please think for yourself about how to solve . Hint: first reduce A to upper or lower triangular form, then to diagonal form, and finally to the identity matrix.

The following system of linear equations is called a homogeneous linear system. A homogeneous linear system is one whose constant terms are all zero (b is the zero vector 0) — that is, a system of the form

$$\begin{cases} x_1 - x_2 = 0 \\ 2x_1 + 2x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & -1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_0$$

As shown in Figure 14, both lines pass through the origin; the intersection of the two lines is the solution to the system of linear equations above.

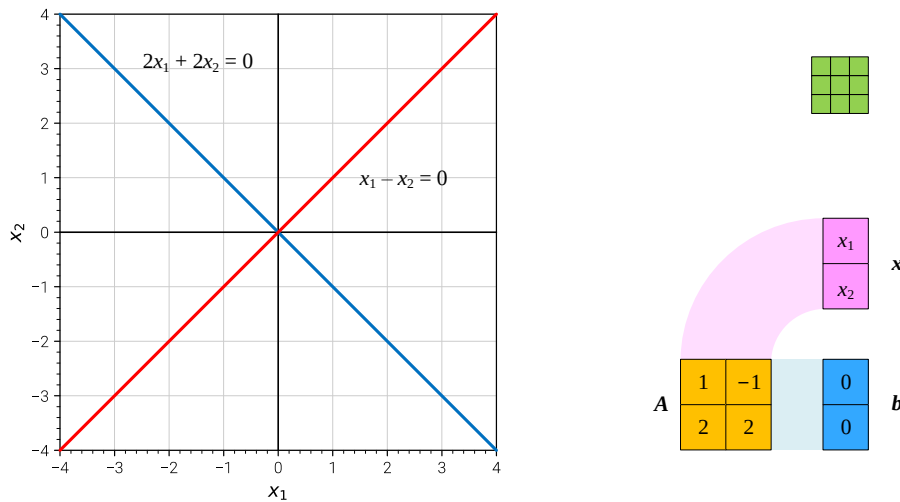


Figure 14. The Matrix A Has Nonzero Determinant, $Ax = 0$



LA_07_01_04.ipynb visualizes the solution of the system of linear equations by plotting contour lines; please study it on your own.

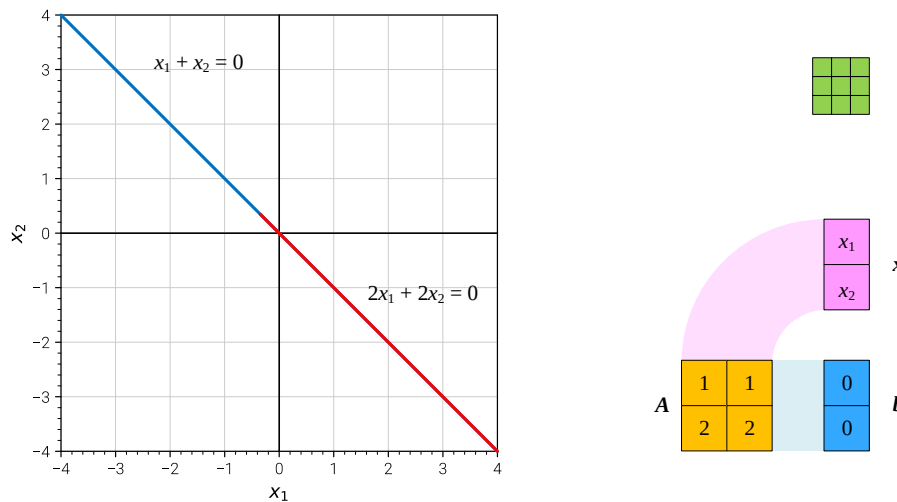
General Square Matrices with Zero Determinant

In the example below, the determinant of the matrix A is 0,

$$\begin{cases} x_1 + x_2 = 0 \\ 2x_1 + 2x_2 = 0 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ 0 \end{bmatrix}}_b$$

Clearly, the matrix A has no inverse.

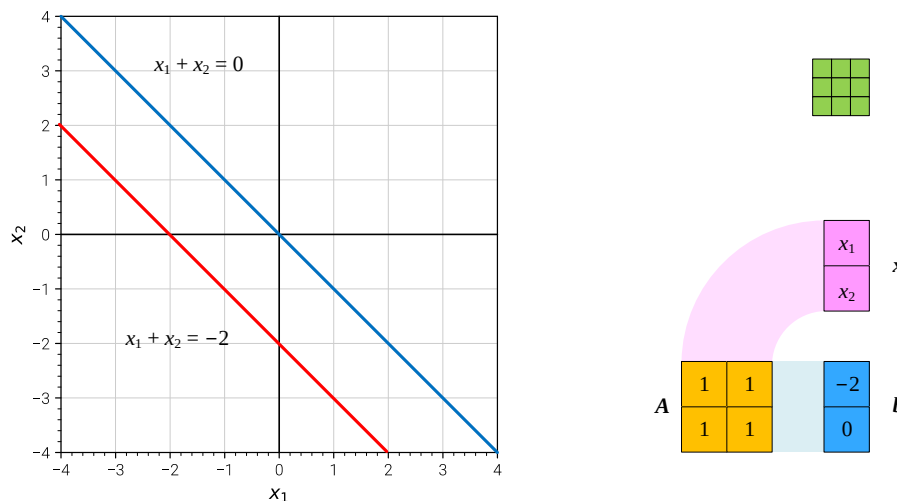
As shown in Figure 15, the two diagonal lines representing the equations above coincide, which means the system has infinitely many solutions.

Figure 15. The Matrix A Has Zero Determinant, $Ax = b$ – Infinitely Many Solutions

In the example below, the determinant of the matrix A is also 0,

$$\begin{cases} x_1 + x_2 = 0 \\ x_1 + x_2 = -2 \end{cases} \Rightarrow \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}}_x = \underbrace{\begin{bmatrix} 0 \\ -2 \end{bmatrix}}_b$$

As shown in Figure 16, the two diagonal lines representing the equations above are parallel, which means the system has no solution.

Figure 16. The Matrix A Has Zero Determinant, $Ax = b$ – No Solution

? Is there another way to interpret ?

Do you still remember the third perspective on matrix multiplication? Let's use this perspective to expand 's matrix multiplication

$$\begin{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} & \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \end{bmatrix}$$

$\begin{matrix} a_1 & a_2 & x & a_1 & a_2 & b \end{matrix}$

The equation above is simply a linear combination. You've probably already noticed that a_1 and a_2 lie on the same line (through the origin), meaning they are linearly dependent. However, b is clearly not on this line!

This is a brand-new perspective that helps us understand systems of linear equations! This is the topic of the next section.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Write the following system of linear equations in matrix-multiplication form.

▶
$$\begin{cases} x_1 + x_2 = 4 \\ 2x_1 + 4x_2 = 12 \end{cases}$$

▶
$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 3x_1 + x_3 = 2 \\ x_1 + 2x_2 + x_3 = 3 \end{cases}$$

▶
$$\begin{cases} x_1 + x_2 = 2 \\ x_1 - x_3 = 1 \\ x_2 - x_3 = 0 \end{cases}$$

Q2. Please write Python code to solve the system of linear equations from the previous question.

Q3. A linear equation in three variables corresponds to a plane in three-dimensional space; please think about what form of equation corresponds to the following figure.

